

Compile this file with XeLaTeX or LuaLaTeX to allow direct Chinese input.

Thermodynamics and Statistical Physics

Spring 2026

Homework 10

Issued: May 31, 2026

Due: **June 8, 2026, before class**

Student Information / 学生信息

Name / 姓名: Xu Chen
Student ID / 学号: 523110910017
Email: xuchen2005@sjtu.edu.cn

Show all essential steps in your derivations.

Reconstructing the van der Waals free energy

Historically, the van der Waals model was introduced through the equation of state

$$P(T, v) = \frac{T}{v - b} - \frac{a}{v^2},$$

where $v = V/N$ is the volume per particle, and $a, b > 0$ are constants. In this problem, we reconstruct the corresponding Helmholtz free energy per particle $f(T, v)$.

Recall the thermodynamic relations

$$P = - \left(\frac{\partial f}{\partial v} \right)_T, \quad s = - \left(\frac{\partial f}{\partial T} \right)_v, \quad u = f + Ts,$$

and

$$c_V = \left(\frac{\partial u}{\partial T} \right)_v = -T \left(\frac{\partial^2 f}{\partial T^2} \right)_v.$$

Assume that

$$c_V = \frac{3}{2},$$

and that in the limit $a \rightarrow 0, b \rightarrow 0$, the free energy reduces to the ideal-gas result

$$f_{\text{IG}}(T, v) = -T \left[\log \left(\frac{v}{\lambda_T^3} \right) + 1 \right], \quad \lambda_T = \frac{\ell}{\sqrt{T}},$$

where ℓ is a constant.

Show that these assumptions determine the van der Waals Helmholtz free energy up to the given normalization as

$$f(T, v) = -T \left[\log \left(\frac{v - b}{\lambda_T^3} \right) + 1 \right] - \frac{a}{v}.$$

Solution

We begin by using the thermodynamic relation $P = - \left(\frac{\partial f}{\partial v} \right)_T$. Given the van der Waals equation of state, we can integrate with respect to v at constant T :

$$\begin{aligned} \left(\frac{\partial f}{\partial v} \right)_T &= -\frac{T}{v - b} + \frac{a}{v^2} \\ f(T, v) &= \int \left(-\frac{T}{v - b} + \frac{a}{v^2} \right) dv \\ f(T, v) &= -T \log(v - b) - \frac{a}{v} + A(T) \end{aligned}$$

where $A(T)$ is a function of temperature only, arising as the constant of integration.

To determine $A(T)$, we use the condition for the specific heat $c_V = 3/2$. We know that

$c_V = -T \left(\frac{\partial^2 f}{\partial T^2} \right)_v$. Let us differentiate our expression for $f(T, v)$:

$$\begin{aligned} \left(\frac{\partial f}{\partial T} \right)_v &= -\log(v - b) + A'(T) \\ \left(\frac{\partial^2 f}{\partial T^2} \right)_v &= A''(T) \end{aligned}$$

Substitute this into the expression for c_V :

$$-TA''(T) = \frac{3}{2} \implies A''(T) = -\frac{3}{2T}$$

Integrating twice with respect to T yields:

$$\begin{aligned} A'(T) &= -\frac{3}{2} \log T + C_1 \\ A(T) &= -\frac{3}{2} T \log T + \frac{3}{2} T + C_1 T + C_2 \end{aligned}$$

To match the ideal gas limit, it is convenient to rewrite the logarithmic term using $\lambda_T = \ell/\sqrt{T}$. Note that $\log(\lambda_T^{-3}) = \log\left(\frac{T^{3/2}}{\ell^3}\right) = \frac{3}{2} \log T - 3 \log \ell$. Thus, we can rewrite $A(T)$ in the form:

$$A(T) = -T \log(\lambda_T^{-3}) + c_3 T + C_2$$

where c_3 regroups the remaining constants. Now we assemble $f(T, v)$:

$$f(T, v) = -T \log(v - b) - \frac{a}{v} - T \log(\lambda_T^{-3}) + c_3 T + C_2 = -T \left[\log\left(\frac{v - b}{\lambda_T^3}\right) - c_3 \right] - \frac{a}{v} + C_2$$

We are given that in the limit $a \rightarrow 0$ and $b \rightarrow 0$, $f(T, v)$ must reduce to the ideal gas free energy:

$$f_{\text{IG}}(T, v) = -T \left[\log\left(\frac{v}{\lambda_T^3}\right) + 1 \right]$$

Taking the limit of our reconstructed $f(T, v)$:

$$\lim_{a, b \rightarrow 0} f(T, v) = -T \left[\log\left(\frac{v}{\lambda_T^3}\right) - c_3 \right] + C_2$$

Matching this with $f_{\text{IG}}(T, v)$ requires $c_3 = -1$ and $C_2 = 0$. Therefore, $A(T) = -T \log(\lambda_T^{-3}) - T$.

Substituting this back gives the full van der Waals Helmholtz free energy:

$$f(T, v) = -T \left[\log\left(\frac{v - b}{\lambda_T^3}\right) + 1 \right] - \frac{a}{v}$$

Recovering $u(s, v)$ from $f(T, v)$

Consider a homogeneous van der Waals fluid with Helmholtz free energy per particle

$$f(T, v) = -T \left[\log \left(\frac{v-b}{\lambda_T^3} \right) + 1 \right] - \frac{a}{v}, \quad \lambda_T = \frac{\ell}{\sqrt{T}},$$

where $v = V/N$, and a, b, ℓ are constants.

Recall that the entropy per particle is

$$s(T, v) = - \left(\frac{\partial f}{\partial T} \right)_v,$$

and that the internal energy as a function of s and v is obtained by the inverse Legendre transform

$$u(s, v) = \max_T [f(T, v) + Ts].$$

- (a) Compute $s(T, v)$.
- (b) Solve the stationarity condition for the maximizing temperature $T_*(s, v)$.
- (c) Show that the stationary point is a maximum.
- (d) Hence show that

$$u(s, v) = \frac{3}{2} \left(\frac{\ell^3}{v-b} \right)^{2/3} \exp \left[\frac{2}{3} \left(s - \frac{5}{2} \right) \right] - \frac{a}{v}.$$

Solution

- (a) We need to compute $s(T, v) = - \left(\frac{\partial f}{\partial T} \right)_v$. First, let us expand the terms in $f(T, v)$ involving T . Since $\lambda_T^3 = \ell^3 T^{-3/2}$, we have:

$$f(T, v) = -T \log(v-b) + T \log(\ell^3 T^{-3/2}) - T - \frac{a}{v} = -T \log(v-b) + 3T \log \ell - \frac{3}{2} T \log T - T - \frac{a}{v}$$

Taking the derivative with respect to T at constant v :

$$\begin{aligned} \left(\frac{\partial f}{\partial T} \right)_v &= -\log(v-b) + 3 \log \ell - \frac{3}{2} (\log T + 1) - 1 \\ &= -\log(v-b) + \log(\ell^3 T^{-3/2}) - \frac{3}{2} - 1 \\ &= -[\log(v-b) - \log(\lambda_T^3)] - \frac{5}{2} = -\log \left(\frac{v-b}{\lambda_T^3} \right) - \frac{5}{2} \end{aligned}$$

Thus, the entropy is:

$$s(T, v) = \log \left(\frac{v-b}{\lambda_T^3} \right) + \frac{5}{2}$$

- (b) We find $T_*(s, v)$ by solving the stationarity condition for $L(T) = f(T, v) + Ts$.

$$\frac{\partial L}{\partial T} = \left(\frac{\partial f}{\partial T} \right)_v + s = 0 \implies s(T_*, v) = s$$

Substitute the expression for entropy from part (a):

$$\log\left(\frac{v-b}{\ell^3 T_*^{-3/2}}\right) + \frac{5}{2} = s$$

Rearranging to solve for T_* :

$$\begin{aligned}\log(v-b) - \log(\ell^3) + \frac{3}{2} \log T_* &= s - \frac{5}{2} \\ \frac{3}{2} \log T_* &= s - \frac{5}{2} - \log\left(\frac{v-b}{\ell^3}\right) \\ T_*^{3/2} &= \left(\frac{\ell^3}{v-b}\right) \exp\left(s - \frac{5}{2}\right) \\ T_*(s, v) &= \left(\frac{\ell^3}{v-b}\right)^{2/3} \exp\left[\frac{2}{3}\left(s - \frac{5}{2}\right)\right]\end{aligned}$$

- (c) To check that this stationary point is a maximum, we evaluate the second derivative of $L(T)$ with respect to T :

$$\frac{\partial^2 L}{\partial T^2} = \left(\frac{\partial^2 f}{\partial T^2}\right)_v = -\frac{c_V}{T}$$

We are given that $c_V = 3/2 > 0$, and temperature must be strictly positive ($T > 0$). Therefore, $\frac{\partial^2 L}{\partial T^2} < 0$, confirming that T_* gives a maximum.

- (d) The internal energy is $u(s, v) = f(T_*, v) + T_* s$. We can simplify the expression for $f(T, v)$ using the entropy derived in part (a). Notice that:

$$f(T, v) = -T \left[s(T, v) - \frac{5}{2} + 1 \right] - \frac{a}{v} = -T s(T, v) + \frac{3}{2} T - \frac{a}{v}$$

Evaluating this at the maximizing temperature T_* (where $s(T_*, v) = s$):

$$u(s, v) = \left(-T_* s + \frac{3}{2} T_* - \frac{a}{v}\right) + T_* s = \frac{3}{2} T_* - \frac{a}{v}$$

Finally, substitute the explicit formula for T_* derived in part (b):

$$u(s, v) = \frac{3}{2} \left(\frac{\ell^3}{v-b}\right)^{2/3} \exp\left[\frac{2}{3}\left(s - \frac{5}{2}\right)\right] - \frac{a}{v}$$

Energy jump, latent heat, and volume jump

Consider liquid–vapor coexistence at a fixed temperature $T < T_c$. Let

$$\Delta v = v_{\text{vap}}^{\text{coex}}(T) - v_{\ell}^{\text{coex}}(T), \quad \Delta s = s_{\text{vap}}^{\text{coex}}(T) - s_{\ell}^{\text{coex}}(T),$$

and define the latent heat per particle by

$$\ell(T) = T \Delta s.$$

Let

$$\Delta u = u_{\text{vap}}^{\text{coex}}(T) - u_{\ell}^{\text{coex}}(T)$$

be the jump in internal energy per particle.

- (a) Use the equality of branch Gibbs free energies at coexistence,

$$g_{\ell}(T, P_{\text{coex}}) = g_{\text{vap}}(T, P_{\text{coex}}),$$

to show that

$$\Delta f + P_{\text{coex}} \Delta v = 0.$$

- (b) Using $f = u - Ts$, show that

$$\Delta u = T \Delta s - P_{\text{coex}} \Delta v.$$

- (c) Conclude that

$$\Delta u = \ell(T) - P_{\text{coex}}(T) \Delta v.$$

Explain why this relation allows the jump in internal energy to be inferred from the latent heat and the volume jump.

Solution

- (a) The Gibbs free energy per particle is defined as $g = f + Pv$. At phase coexistence, the branch Gibbs free energies of the liquid and vapor phases are equal:

$$g_{\ell}(T, P_{\text{coex}}) = g_{\text{vap}}(T, P_{\text{coex}})$$

Substitute the definition of g :

$$f_{\ell} + P_{\text{coex}} v_{\ell} = f_{\text{vap}} + P_{\text{coex}} v_{\text{vap}}$$

Rearranging the terms yields:

$$(f_{\text{vap}} - f_{\ell}) + P_{\text{coex}}(v_{\text{vap}} - v_{\ell}) = 0$$

Using the Δ notation, this is exactly:

$$\Delta f + P_{\text{coex}} \Delta v = 0$$

- (b) The definition of the Helmholtz free energy is $f = u - Ts$. For an isothermal process (both phases are at the same temperature T), taking the difference between the vapor and liquid phases gives:

$$f_{\text{vap}} - f_{\ell} = (u_{\text{vap}} - u_{\ell}) - T(s_{\text{vap}} - s_{\ell}) \implies \Delta f = \Delta u - T\Delta s$$

Substitute this expression for Δf into the result from part (a):

$$(\Delta u - T\Delta s) + P_{\text{coex}}\Delta v = 0$$

Rearranging for Δu yields:

$$\Delta u = T\Delta s - P_{\text{coex}}\Delta v$$

- (c) By definition, the latent heat per particle is $\ell(T) = T\Delta s$. Substituting this into the result from part (b) gives:

$$\Delta u = \ell(T) - P_{\text{coex}}(T)\Delta v$$

Explanation: This relation elegantly decomposes the jump in internal energy into two distinct physical contributions. The term $\ell(T)$ represents the total heat energy added to the system to drive the phase transition. However, not all of this heat goes into the internal energy; as the substance vaporizes, it expands by Δv against the constant surrounding coexistence pressure P_{coex} , thus doing mechanical work $P_{\text{coex}}\Delta v$. The internal energy jump is simply the heat added minus the work done by the system. This allows Δu to be inferred purely from macroscopic, experimentally measurable quantities (latent heat, coexistence pressure, and volume jump).

Near-critical spinodal and coexistence curves of the van der Waals fluid

In the lecture notes, we developed the general construction of phase coexistence using branch Gibbs free energies, Maxwell's equal-area rule, and the double-tangent construction. In this problem, we apply that theory to the van der Waals equation near the critical point. The goal is to compute the spinodal and binodal curves systematically in a small parameter and to extract the mean-field critical behavior.

The van der Waals equation of state is

$$P = \frac{T}{v - b} - \frac{a}{v^2},$$

with critical point

$$v_c = 3b, \quad T_c = \frac{8a}{27b}, \quad P_c = \frac{a}{27b^2}.$$

Introduce the reduced variables

$$\theta = \frac{T - T_c}{T_c}, \quad \pi = \frac{P - P_c}{P_c}, \quad \varphi = \frac{v - v_c}{v_c}.$$

For the rest of the problem we focus on subcritical isotherms, $T < T_c$, and introduce the positive small parameter

$$\epsilon = \sqrt{-\theta} = \left(\frac{T_c - T}{T_c} \right)^{1/2}.$$

Near the critical point the relevant volume deviations scale as

$$\varphi = O(\epsilon).$$

We therefore introduce the scaled volume variable

$$\varphi = \epsilon y.$$

Thus the critical point is approached by taking

$$\epsilon \rightarrow 0$$

with $y = O(1)$.

(a) Show that the van der Waals equation of state can be written exactly as

$$1 + \pi = \frac{8(1 + \theta)}{2 + 3\varphi} - \frac{3}{(1 + \varphi)^2}.$$

After setting

$$\theta = -\epsilon^2, \quad \varphi = \epsilon y,$$

show that this becomes

$$1 + \pi = \frac{8(1 - \epsilon^2)}{2 + 3\epsilon y} - \frac{3}{(1 + \epsilon y)^2}.$$

Expand near the critical point and show that

$$\pi = -4\epsilon^2 + \epsilon^3 \left(6y - \frac{3}{2}y^3 \right) + \epsilon^4 \left(-9y^2 + \frac{21}{4}y^4 \right) + O(\epsilon^5).$$

This expansion shows that the pressure deviation in the critical scaling region begins at order ϵ^2 , while the first y -dependent terms begin at order ϵ^3 . This reflects the fact that, at the critical point, both the linear and quadratic dependence of π on φ vanish.

Before introducing the scaled variable y , the spinodal condition is

$$\left(\frac{\partial \pi}{\partial \varphi} \right)_{\theta} = 0.$$

At the critical point, the spinodal point also becomes an inflection point of the critical isotherm, so it satisfies the additional condition

$$\left(\frac{\partial^2 \pi}{\partial \varphi^2} \right)_{\theta} = 0.$$

Verify these two conditions directly from the exact reduced equation of state at

$$\theta = 0, \quad \varphi = 0.$$

- (b) We now determine the spinodal curve near the critical point. Since $\varphi = \epsilon y$, and ϵ is fixed along a given isotherm, the spinodal condition is equivalent to

$$\left(\frac{\partial \pi}{\partial y} \right)_{\epsilon} = 0.$$

Using the expansion from part (a), show that this condition gives

$$0 = 6 - \frac{9}{2}y^2 + \epsilon (21y^3 - 18y) + O(\epsilon^2).$$

Now assume an expansion

$$y_{\text{sp}} = y_0 + y_1\epsilon + O(\epsilon^2).$$

Insert this expansion into the spinodal condition and compare the two lowest orders in ϵ . Deduce the two spinodal branches:

$$\varphi_{\text{sp}}^{\ell} = -\frac{2}{\sqrt{3}}\epsilon + \frac{10}{9}\epsilon^2 + O(\epsilon^3),$$

and

$$\varphi_{\text{sp}}^{\text{vap}} = \frac{2}{\sqrt{3}}\epsilon + \frac{10}{9}\epsilon^2 + O(\epsilon^3).$$

Verify that these two branches merge at the critical point.

(c) Compute the corresponding spinodal pressures and show that

$$\pi_{\text{sp}}^\ell = -4\epsilon^2 - \frac{8\sqrt{3}}{3}\epsilon^3 - \frac{8}{3}\epsilon^4 + O(\epsilon^5),$$

and

$$\pi_{\text{sp}}^{\text{vap}} = -4\epsilon^2 + \frac{8\sqrt{3}}{3}\epsilon^3 - \frac{8}{3}\epsilon^4 + O(\epsilon^5).$$

(d) We now determine the leading coexistence volumes. Since the coexistence volumes lie in the same critical scaling window as the spinodal volumes, write

$$y_\ell = y_{\ell,0} + O(\epsilon), \quad y_{\text{vap}} = y_{\text{vap},0} + O(\epsilon).$$

Using the scaled equation of state from part (a), for $y = O(1)$ we have

$$\pi = -4\epsilon^2 + O(\epsilon^3).$$

We may therefore write the coexistence pressure in the form

$$\pi_{\text{coex}} = -4\epsilon^2 + \epsilon^3 q + O(\epsilon^4),$$

where q is to be determined.

The two coexistence endpoints must have the same pressure:

$$\pi(\epsilon, y_\ell) = \pi(\epsilon, y_{\text{vap}}) = \pi_{\text{coex}}.$$

Using the scaled equation of state and comparing the first nontrivial order, show that

$$6y_{\ell,0} - \frac{3}{2}y_{\ell,0}^3 = q, \quad 6y_{\text{vap},0} - \frac{3}{2}y_{\text{vap},0}^3 = q.$$

Next use Maxwell's equal-area rule,

$$\int_{\varphi_\ell}^{\varphi_{\text{vap}}} [\pi - \pi_{\text{coex}}] d\varphi = 0, \quad \varphi_\ell = \epsilon y_\ell, \quad \varphi_{\text{vap}} = \epsilon y_{\text{vap}}.$$

Since $\varphi = \epsilon y$, this is equivalent to

$$\int_{y_\ell}^{y_{\text{vap}}} [\pi(\epsilon, y) - \pi_{\text{coex}}] dy = 0.$$

At leading order this gives

$$\int_{y_{\ell,0}}^{y_{\text{vap},0}} \left[6y - \frac{3}{2}y^3 - q \right] dy = 0.$$

Together with the two endpoint conditions, show that

$$q = 0, \quad y_{\ell,0} = -2, \quad y_{\text{vap},0} = 2.$$

(e) Now compute the next correction to the coexistence volumes and pressure. Write

$$y_\ell = -2 + a\epsilon + O(\epsilon^2), \quad y_{\text{vap}} = 2 + b\epsilon + O(\epsilon^2),$$

and

$$\pi_{\text{coex}} = -4\epsilon^2 + \epsilon^4 r + O(\epsilon^5),$$

where we have used the result $q = 0$ from the previous part.

Insert these expansions into the endpoint pressure conditions and compare terms through order ϵ^4 . Show that

$$r = 48 - 12a, \quad r = 48 - 12b.$$

Hence

$$a = b.$$

Because the leading-order integrand vanishes at the endpoints $y = \pm 2$, the endpoint shifts do not contribute at this order. Use the next-order term in Maxwell's equal-area rule to show that

$$\int_{-2}^2 \left[\frac{21}{4}y^4 - 9y^2 - r \right] dy = 0.$$

Determine r , and then determine a and b . Conclude that

$$a = b = \frac{18}{5}, \quad r = \frac{24}{5}.$$

(f) Conclude that the coexistence volumes are

$$\varphi_\ell = -2\epsilon + \frac{18}{5}\epsilon^2 + O(\epsilon^3),$$

and

$$\varphi_{\text{vap}} = 2\epsilon + \frac{18}{5}\epsilon^2 + O(\epsilon^3).$$

Therefore the $O(\epsilon^2)$ correction cancels in the volume difference, and

$$\varphi_{\text{vap}} - \varphi_\ell = 4\epsilon + O(\epsilon^3).$$

Equivalently,

$$v_{\text{vap}}^{\text{coex}}(T) - v_{\ell}^{\text{coex}}(T) = 4v_c \left(\frac{T_c - T}{T_c} \right)^{1/2} + O\left((T_c - T)^{3/2}\right).$$

Thus the order parameter vanishes with the mean-field exponent $1/2$.

(g) Show that the coexistence pressure is

$$\pi_{\text{coex}} = -4\epsilon^2 + \frac{24}{5}\epsilon^4 + O(\epsilon^5).$$

Equivalently,

$$P_{\text{coex}}(T) = P_c \left[1 + 4 \frac{T - T_c}{T_c} + \frac{24}{5} \left(\frac{T - T_c}{T_c} \right)^2 + \dots \right].$$

This gives the near-critical binodal curve in the T - P plane.

Solution

- (a) Starting with the equation of state $P = \frac{T}{v-b} - \frac{a}{v^2}$, we express it in terms of reduced variables $1 + \pi = P/P_c$, $1 + \theta = T/T_c$, and $1 + \varphi = v/v_c$. Factoring out P_c :

$$1 + \pi = \frac{1}{P_c} \left(\frac{T_c(1 + \theta)}{v_c(1 + \varphi) - b} - \frac{a}{v_c^2(1 + \varphi)^2} \right)$$

Substitute $v_c = 3b$, $T_c = \frac{8a}{27b}$, and $1/P_c = \frac{27b^2}{a}$:

$$\begin{aligned} 1 + \pi &= \frac{27b^2}{a} \left(\frac{\frac{8a}{27b}(1 + \theta)}{3b(1 + \varphi) - b} - \frac{a}{9b^2(1 + \varphi)^2} \right) \\ &= \frac{8(1 + \theta)}{3(1 + \varphi) - 1} - \frac{3}{(1 + \varphi)^2} \\ &= \frac{8(1 + \theta)}{2 + 3\varphi} - \frac{3}{(1 + \varphi)^2} \end{aligned}$$

Now insert the critical scaling relations $\theta = -\epsilon^2$ and $\varphi = \epsilon y$:

$$1 + \pi = \frac{8(1 - \epsilon^2)}{2 + 3\epsilon y} - \frac{3}{(1 + \epsilon y)^2}$$

We expand this expression for small ϵ using $(1 + x)^{-n} \approx 1 - nx + \frac{n(n+1)}{2}x^2 - \dots$:

$$\begin{aligned} \frac{8(1 - \epsilon^2)}{2(1 + \frac{3}{2}\epsilon y)} &= 4(1 - \epsilon^2) \left[1 - \frac{3}{2}\epsilon y + \frac{9}{4}\epsilon^2 y^2 - \frac{27}{8}\epsilon^3 y^3 + \frac{81}{16}\epsilon^4 y^4 + \dots \right] \\ &= 4 \left[1 - \frac{3}{2}\epsilon y + \epsilon^2 \left(\frac{9}{4}y^2 - 1 \right) + \epsilon^3 \left(-\frac{27}{8}y^3 + \frac{3}{2}y \right) + \epsilon^4 \left(\frac{81}{16}y^4 - \frac{9}{4}y^2 \right) \right] \\ &= 4 - 6\epsilon y + \epsilon^2(9y^2 - 4) + \epsilon^3 \left(-\frac{27}{2}y^3 + 6y \right) + \epsilon^4 \left(\frac{81}{4}y^4 - 9y^2 \right) \\ \frac{3}{(1 + \epsilon y)^2} &= 3 \left[1 - 2\epsilon y + 3\epsilon^2 y^2 - 4\epsilon^3 y^3 + 5\epsilon^4 y^4 + \dots \right] \\ &= 3 - 6\epsilon y + 9\epsilon^2 y^2 - 12\epsilon^3 y^3 + 15\epsilon^4 y^4 \end{aligned}$$

Subtracting the second from the first gives $1 + \pi$:

$$\begin{aligned} 1 + \pi &= (4 - 3) + (-6 + 6)\epsilon y + \epsilon^2(9y^2 - 4 - 9y^2) \\ &\quad + \epsilon^3 \left(-\frac{27}{2}y^3 + 6y + 12y^3 \right) + \epsilon^4 \left(\frac{81}{4}y^4 - 9y^2 - 15y^4 \right) \\ &= 1 - 4\epsilon^2 + \epsilon^3 \left(6y - \frac{3}{2}y^3 \right) + \epsilon^4 \left(-9y^2 + \frac{21}{4}y^4 \right) + O(\epsilon^5) \end{aligned}$$

Subtracting 1 verifies the given expansion for π .

To verify the critical point conditions at $\theta = 0, \varphi = 0$, we differentiate the exact relation $\pi(\varphi, \theta)$:

$$\begin{aligned} \left(\frac{\partial \pi}{\partial \varphi} \right)_{\theta} &= -\frac{24(1 + \theta)}{(2 + 3\varphi)^2} + \frac{6}{(1 + \varphi)^3} \xrightarrow{\theta=0, \varphi=0} -\frac{24}{4} + 6 = -6 + 6 = 0 \\ \left(\frac{\partial^2 \pi}{\partial \varphi^2} \right)_{\theta} &= \frac{144(1 + \theta)}{(2 + 3\varphi)^3} - \frac{18}{(1 + \varphi)^4} \xrightarrow{\theta=0, \varphi=0} \frac{144}{8} - 18 = 18 - 18 = 0 \end{aligned}$$

Both the spinodal and the inflection conditions hold exactly.

(b) The spinodal condition is $\left(\frac{\partial\pi}{\partial y}\right)_\epsilon = 0$. Differentiating the expansion from (a):

$$\left(\frac{\partial\pi}{\partial y}\right)_\epsilon = \epsilon^3 \left(6 - \frac{9}{2}y^2\right) + \epsilon^4 (-18y + 21y^3) + O(\epsilon^5) = 0$$

Dividing out ϵ^3 , we get:

$$0 = 6 - \frac{9}{2}y^2 + \epsilon (21y^3 - 18y) + O(\epsilon^2)$$

Insert the expansion $y_{\text{sp}} = y_0 + y_1\epsilon$:

$$6 - \frac{9}{2}(y_0^2 + 2\epsilon y_0 y_1) + \epsilon(21y_0^3 - 18y_0) = 0$$

Matching order ϵ^0 :

$$6 - \frac{9}{2}y_0^2 = 0 \implies y_0^2 = \frac{4}{3} \implies y_0 = \pm \frac{2}{\sqrt{3}}$$

Matching order ϵ^1 :

$$-9y_0 y_1 + 21y_0^3 - 18y_0 = 0 \implies y_1 = \frac{21y_0^2 - 18}{9} = \frac{21(4/3) - 18}{9} = \frac{28 - 18}{9} = \frac{10}{9}$$

Since $\varphi = \epsilon y$, we multiply by ϵ to get the two branches:

$$\begin{aligned}\varphi_{\text{sp}}^\ell &= -\frac{2}{\sqrt{3}}\epsilon + \frac{10}{9}\epsilon^2 + O(\epsilon^3) \\ \varphi_{\text{sp}}^{\text{vap}} &= \frac{2}{\sqrt{3}}\epsilon + \frac{10}{9}\epsilon^2 + O(\epsilon^3)\end{aligned}$$

As $\epsilon \rightarrow 0$ (approaching T_c), both $\varphi_{\text{sp}}^\ell \rightarrow 0$ and $\varphi_{\text{sp}}^{\text{vap}} \rightarrow 0$, merging at the critical point.

(c) Substitute $y_{\text{sp}} = y_0 + y_1\epsilon$ back into $\pi(y)$. For the $O(\epsilon^3)$ term:

$$6y_0 - \frac{3}{2}y_0^3 = y_0 \left(6 - \frac{3}{2}y_0^2\right) = \left(\pm \frac{2}{\sqrt{3}}\right) \left(6 - \frac{3}{2} \frac{4}{3}\right) = \left(\pm \frac{2}{\sqrt{3}}\right) (4) = \pm \frac{8}{\sqrt{3}} = \pm \frac{8\sqrt{3}}{3}$$

For the $O(\epsilon^4)$ term, note the contribution from y_1 evaluates to $y_1(6 - \frac{9}{2}y_0^2) = 0$ because y_0 is the root. Thus, we only need $-9y_0^2 + \frac{21}{4}y_0^4$:

$$-9 \left(\frac{4}{3}\right) + \frac{21}{4} \left(\frac{16}{9}\right) = -12 + \frac{21 \times 4}{9} = -12 + \frac{28}{3} = -12 + 9.333 = -\frac{8}{3}$$

Thus, the spinodal pressures are:

$$\begin{aligned}\pi_{\text{sp}}^\ell(y_0 < 0) &= -4\epsilon^2 - \frac{8\sqrt{3}}{3}\epsilon^3 - \frac{8}{3}\epsilon^4 + O(\epsilon^5) \\ \pi_{\text{sp}}^{\text{vap}}(y_0 > 0) &= -4\epsilon^2 + \frac{8\sqrt{3}}{3}\epsilon^3 - \frac{8}{3}\epsilon^4 + O(\epsilon^5)\end{aligned}$$

- (d) The coexistence pressure is $\pi_{\text{coex}} = -4\epsilon^2 + \epsilon^3 q$. Matching endpoints $\pi(y_{\ell,0}) = \pi(y_{\text{vap},0}) = \pi_{\text{coex}}$ at $O(\epsilon^3)$:

$$6y_{\ell,0} - \frac{3}{2}y_{\ell,0}^3 = q \quad \text{and} \quad 6y_{\text{vap},0} - \frac{3}{2}y_{\text{vap},0}^3 = q$$

Maxwell's equal-area rule requires:

$$\int_{y_{\ell,0}}^{y_{\text{vap},0}} \left[6y - \frac{3}{2}y^3 - q \right] dy = 0$$

Because $f(y) = 6y - \frac{3}{2}y^3$ is an odd function, the integral of $f(y) - q$ between its roots vanishes if and only if $q = 0$ and the roots are symmetric ($y_{\ell,0} = -y_{\text{vap},0}$). Setting $q = 0$:

$$6y_0 - \frac{3}{2}y_0^3 = 0 \implies y_0 \left(6 - \frac{3}{2}y_0^2 \right) = 0$$

The non-trivial solutions are $y_0^2 = 4 \implies y_0 = \pm 2$. Thus, $q = 0$, $y_{\ell,0} = -2$, and $y_{\text{vap},0} = 2$.

- (e) Next, we incorporate the next order shifts $y_{\ell} = -2 + a\epsilon$, $y_{\text{vap}} = 2 + b\epsilon$, and $\pi_{\text{coex}} = -4\epsilon^2 + \epsilon^4 r$. We evaluate π at $y = \pm 2 + \delta\epsilon$ up to order ϵ^4 . The $O(\epsilon^3)$ term evaluated with the shift is:

$$\epsilon^3 \left[6(\pm 2 + \delta\epsilon) - \frac{3}{2}(\pm 2 + \delta\epsilon)^3 \right] = \epsilon^3 [(\pm 12 \mp 12) + (6\delta - 18\delta)\epsilon] = -12\delta\epsilon^4$$

The $O(\epsilon^4)$ term evaluated at $y_0 = \pm 2$ is:

$$\epsilon^4 \left[-9(\pm 2)^2 + \frac{21}{4}(\pm 2)^4 \right] = \epsilon^4 \left[-36 + \frac{21}{4}(16) \right] = \epsilon^4(-36 + 84) = 48\epsilon^4$$

Matching $\pi(y_{\ell})$ and $\pi(y_{\text{vap}})$ to π_{coex} at order ϵ^4 :

$$-12a + 48 = r \quad \text{and} \quad -12b + 48 = r$$

This forces $a = b$. Now apply Maxwell's rule at the next order. The endpoint shift contributions vanish because the leading integrand is identically zero at $y = \pm 2$. We integrate the $O(\epsilon^4)$ terms from -2 to 2 :

$$\int_{-2}^2 \left[\frac{21}{4}y^4 - 9y^2 - r \right] dy = 0$$

Evaluate the integral:

$$\left[\frac{21}{20}y^5 - 3y^3 - ry \right]_{-2}^2 = 2 \left(\frac{48}{5} - 2r \right) = 0$$

This yields $2r = 48/5 \implies r = 24/5$. Substitute r back to find a :

$$12a = 48 - \frac{24}{5} = \frac{216}{5} \implies a = b = \frac{18}{5}$$

(f) Restoring the scale factor ϵ , the coexistence volumes are:

$$\begin{aligned}\varphi_\ell &= -2\epsilon + \frac{18}{5}\epsilon^2 + O(\epsilon^3) \\ \varphi_{\text{vap}} &= 2\epsilon + \frac{18}{5}\epsilon^2 + O(\epsilon^3)\end{aligned}$$

The $O(\epsilon^2)$ correction is perfectly symmetric, so it cancels when taking the volume difference:

$$\varphi_{\text{vap}} - \varphi_\ell = 4\epsilon + O(\epsilon^3)$$

Since $\varphi = (v - v_c)/v_c$ and $\epsilon = \left(\frac{T_c - T}{T_c}\right)^{1/2}$, we obtain:

$$v_{\text{vap}}^{\text{coex}}(T) - v_\ell^{\text{coex}}(T) = 4v_c \left(\frac{T_c - T}{T_c}\right)^{1/2} + O\left((T_c - T)^{3/2}\right)$$

This establishes that the order parameter for the phase transition vanishes exactly with the universal mean-field critical exponent $\beta = 1/2$.

(g) With $q = 0$ and $r = 24/5$, the coexistence pressure up to order ϵ^4 is:

$$\pi_{\text{coex}} = -4\epsilon^2 + \frac{24}{5}\epsilon^4 + O(\epsilon^5)$$

Substitute $\epsilon^2 = \frac{T_c - T}{T_c} = -\theta$ and $\pi_{\text{coex}} = \frac{P_{\text{coex}} - P_c}{P_c}$:

$$P_{\text{coex}}(T) = P_c \left[1 + 4\frac{T - T_c}{T_c} + \frac{24}{5} \left(\frac{T - T_c}{T_c}\right)^2 + \dots \right]$$

This provides the analytic form of the near-critical binodal curve in the T - P plane.